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Studierichting: Informatica

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$$\begin{aligned} & \int \sin^{-2}(x) \cos^{-4}(x) dx \\ &= \int \frac{dx}{\frac{\sin^2(x)}{\cos^2(x)} \cdot \cos^6(x)} = \int \frac{dx}{\tan^2(x) \cos^6(x)} \\ &= \int \frac{\sec^6(x)}{\tan^2(x)} dx = \int \frac{(1 + \tan^2(x))^2 \cdot \sec^2(x)}{\tan^2(x)} dx \\ &= \int \frac{(1 + 2 \tan^2(x) + \tan^4(x)) \sec^2(x)}{\tan^2(x)} dx \\ &= \int \left(\frac{1}{\tan^2(x)} + 2 + \tan^2(x) \right) \sec^2(x) dx \\ & \text{Stel } u = \tan(x) \Rightarrow du = \sec^2(x) dx \\ &= \int (u^{-2} + 2 + u^2) du \\ &= \frac{-1}{\tan(x)} + 2 \tan(x) + \frac{\tan^3(x)}{3} + c \end{aligned}$$

References